

A State-of-the-Art Review on Multimodal Deep Learning for Medical Diagnosis: Integrating Text and Image

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Abstract

Multimodal deep learning has now been recognized as a prominent paradigm in the field of computer-assisted medical diagnosis, which facilitates the integration of diverse data sources such as medical images, text data, physiological signals, and electronic health records. The traditional single-modal medical diagnostic systems tend to fail in handling the complex and coupled aspects of medical information, leading to suboptimal diagnostic performance, low robustness, and poor generalizability across different disease domains. In contrast, multimodal learning systems utilize supplementary information from other modalities to create more abstract feature representations, thus providing better support for more accurate clinical decision-making. This review provides a comprehensive and systematic treatment of recent developments in multimodal deep learning-based medical diagnostics, which considers the published work from 2021 to 2025. The review focuses on six major domains of diseases: Brain and Neurological Disorders, Lung and Respiratory Diseases, Cancer and Oncology, Cardiovascular Diseases, Infectious Diseases, and General Multimodal Diagnostic Frameworks. In these domains, various types of datasets like imaging, text, audio, genetic, and time-series clinical data are considered together with data preprocessing methods, fusion architectures, evaluation metrics, and explainability tools. The results show a significant shift from conventional convolutional neural network-based, and late fusion-based architectures to more advanced models such as transformer-based and attention-driven models, which are adept at aligning representations across different modalities. The growing availability of self-supervised, synthetic data augmentation, and multimodal large language models has significantly improved overall performance, especially in scenarios of limited clinical data availability. However, several challenges still persist in multimodal deep learning, including the unavailability of multimodal datasets, complexity of modality fusion, spatio-temporal misalignment, and the trade-off between interpretability and performance accuracy. This review indicates the potential of multimodal deep learning in improving the accuracy, robustness, and clinical relevance of medical diagnostics in various applications, while also suggesting important research directions required for widespread and interpretable clinical practice.

1. Introduction

Artificial intelligence (AI) and deep learning have progressed at a fast pace in the field of modern healthcare, especially in the context of medical diagnosis, where decision-making involves the effective amalgamation of disparate sources of information. Classical machine learning algorithms conventionally follow a unimodal approach, focusing on imaging, electrocardiography (ECG), or text data, which in turn restricts their ability to comprehend the underlying complexity of human disease. Recent breakthroughs in multimodal learning aim to fill this gap by aggregating complementary sources of data and then learning compact representations. Thus, multimodal AI has emerged as one of the most promising areas in medical diagnostics, risk assessment, and precision medicine for a variety of diseases [92, 93]. Simultaneously, in this emerging landscape of AI applications in the medical domain, there is an increasing need for Explainable AI (XAI) techniques. Comprehensive surveys on XAI have emphasized that clinical AI systems must provide transparent and interpretable reasoning to gain clinician trust and ensure patient safety [123, 121]. The importance of causability and explainability in medical AI has been highlighted as a prerequisite for clinical adoption, particularly in high-stakes diagnostic tasks [125]. Self-supervised learning paradigms have also

started to emerge as potential alternatives to large-scale labeled datasets, offering competitive performance while reducing annotation burden in medical imaging applications [69]. Taken together, these developments represent an increasing consensus that clinical AI must be transparent, reliable, and able to synthesize various types of information about patients.

Although these are important individual contributions, most of the existing literature is still restricted to single-modal setups or particular disease scenarios. Consequently, the literature lacks a comprehensive understanding across domains regarding how multimodal learning performs for varied diseases, characteristics of datasets, and fusion models. Moreover, since individual contributions aim to solve the problem of explainability in a specific setup, there is no need for a comparison of the performance of XAI techniques across multimodal pipelines combining imaging, text, audio, genomics, EHRs, and physiological signals together.

To address these gaps, this paper presents a comprehensive 2021–2025 review of multimodal medical AI, covering six major disease domains:

1. brain and neurological disorders,
2. lung and respiratory diseases,
3. cancer and oncology,
4. cardiovascular disorders,
5. infectious diseases, and
6. general multimodal diagnostic frameworks.

We examine each of the studies in detail, using their dataset characterization, preprocessing techniques, fusion approaches, model designs, explainability methods, and results. In addition to findings within the domain, this study points out some general trends, including the preponderance of imaging-text fusion, attention-based models, and the use of self-supervised multimodal foundations. Secondly, we point out some of the essential challenges that need to be considered, including the unavailability of multimodal datasets, the mismatch of cross-modal data, spatio-temporal inconsistencies, fusion complexity, and the accuracy-explainability tradeoff. By providing a comprehensive overview of multimodal research in various clinical domains, this paper lays out a foundation for the development of future robust, explainable, and clinically-feasible multimodal AI systems for healthcare applications. The broad objective of this work is the development of accurate and interpretable diagnostic models that can fuse the entire range of medical information in healthcare. The paper has the following key contributions:

- Extensive multimodal analysis over six prominent biomedical domains: Brain and Neurological Disorders, Lung/Respiratory, Cancer/Oncology, Cardiovascular Diseases, Infectious Diseases, and General Multimodal Diagnostic Frameworks, spanning from 2021 to 2025, in contrast to previous studies targeting one disease or one modality.
- A unified framework for comparative analysis to evaluate each study based on the type of data used, the preprocessing technique applied, the fusion approach used, the architecture design, the performance level, and the explanation technique. Introduction of a standardized multimodal workflow diagram, synthesizing the common research pipeline across different types of diseases.

- Identification and categorization of 12 key limitations in existing multimodal research, including the scarcity of multimodal datasets, asynchronous data alignment, data fusion complexity, spatio-temporal mismatch, interpretability issues, and clinical generalizability gaps, among others.
- In-depth analysis of the characteristics of datasets, pointing out the relative ease of use or difficulty of working with different types of data (image, text, EHR, audio, sensor), and the impact of modality characteristics on preprocessing, model performance, and fusion.
- Cross-domain performance comparison to illustrate how various disease tasks systematically differ in terms of accuracy, AUROC, F1-score, and calibration for multimodal models.
- Integration of XAI-centric views, contrast studies using Grad-CAM, SHAP, attention maps, or integrated gradients with those not using interpretability—highlighting the increasing need for clinical transparency in multimodal AI.
- Synthesis of future research directions, including scalable data curation approaches, sound cross-institutional fusion frameworks, self-supervised multimodal models, and clinically informed standards of interpretability.

2. Survey Methodology

The systematic review, as illustrated in Fig. 1, adopted a three-step process in screening studies related to multimodal deep learning for the medical diagnosis task, following PRISMA guidelines.

Search Strategy. Articles were retrieved from IEEE Xplore, PubMed, Scopus, Web of Science, and Google Scholar using the following search query combinations: (“multimodal” OR “multi-modal”) AND (“deep learning” OR “neural network”) AND (“medical diagnosis” OR “clinical diagnosis” OR “disease detection”) AND (“text” OR “image” OR “fusion”). The search was restricted to publications from January 2021 to December 2025.

Inclusion Criteria. Studies were included if they: (1) proposed or evaluated a multimodal deep learning model integrating at least two data modalities (e.g., image + text, image + EHR); (2) targeted a medical diagnosis or clinical decision support task; (3) reported quantitative performance metrics; and (4) were published in peer-reviewed journals, conference proceedings, or established preprint repositories.

Exclusion Criteria. Studies were excluded if they: (1) focused exclusively on singlemodality systems; (2) did not provide sufficient technical detail on the fusion methodology; (3) were review articles, editorials, or commentaries; or (4) were duplicate publications.

Screening Process. Beginning with the identification of studies, 350 articles were retrieved from the bibliographic databases. At the identification stage, 167 articles were removed before screening, based on the criteria above, leaving a total of 183 articles for title and abstract screening. At the screening stage, 84 articles were removed mainly due to their focus on individual modalities instead of multimodal systems, resulting in 99 articles for full-text screening. An additional 59 articles were removed due to insufficient explanation of multimodal integration techniques or lack of quantitative evaluation. Finally, 40 articles satisfied all inclusion criteria and were selected for in-depth analysis.

Quality Assessment. Each included study was evaluated across five quality dimensions: (1) clarity of multimodal architecture description, (2) appropriateness of dataset and experimental design, (3) use of established evaluation metrics, (4) presence of ablation studies or modality contribution analysis, and (5) inclusion of explainability or interpretability methods. While a formal quality scoring tool such as QUADAS-2 was not applied due to the heterogeneity of study designs across the six disease domains, these dimensions guided the comparative analysis presented in Sections 5 and 6.

3. Datasets

This section presents the extensive categorization of multimodal medical diagnosis datasets surveyed in this review, as shown in Fig. 2. The datasets are categorized into six main domains of clinical applications: (1) General Multimodal datasets; (2) Brain and Neurological Disorders; (3) Lung and Respiratory Disorders; (4) Cancer and Oncology; (5) Cardiovascular Diseases; and (6) Infectious Diseases.

3.1. Brain and Neurological Diseases Datasets

3.1.1. PPMI Dataset

The PPMI (Parkinson's Progression Markers Initiative) dataset comprises 90 patients' 3D MRI scans and corresponding clinical records. The patients include both healthy controls

and prodromal Parkinson's disease cases. 12 features were selected from 79 extracted clinical dataset features for training based on clinical records, motor symptoms, and family histories [7]. The multimodal nature of the dataset made it easier to train the deep learning models aimed at early detection and classification of Parkinson's disease subtypes.

3.1.2. TADPOLE and ABIDE Dataset

The dataset comprises two multimodal sources: the Alzheimer's Disease Prediction Of Longitudinal Evolution (TADPOLE) dataset and the Autism Brain Imaging Data Exchange (ABIDE) dataset. The TADPOLE dataset, a subset of the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, contained 6 modalities, including MRI, PET, cognitive tests, CSF biomarkers, risk factors, and demographic information. The dataset was collected from 598 subjects classified into normal control (211 patients), stable mild cognitive impairment (275 patients), and Alzheimer's disease (72 patients). The ABIDE dataset contains functional and structural brain imaging data collected to investigate the neurobiological basis of autism. The dataset consists of 539 individuals diagnosed with Autism Spectrum Disorder (ASD) and 573 control subjects (NC) across four modalities. For this study, 871 patients were selected, comprising 468 typical controls and 403 individuals with ASD [8].

3.1.3. BPPV Dataset

The BPPV (Benign Paroxysmal Positional Vertigo) dataset comprises eye movement video and diagnostic data from 518 patients diagnosed with BPPV. The cases were categorized into six BPPV types based on otolith location. The dataset was split into training and testing sets in an 8:2 ratio [9].

3.1.4. Whole Brain Atlas Dataset

The Whole Brain Atlas dataset (published by Harvard Medical School) comprises 1028 pairs of multimodal medical slice images for training, including MRI-T1 and MRI-T2 images (356 pairs), CT and MRI-T2 images (102 pairs), PET and MRI-T2 images (231 pairs), and SPECT and MRI-T2 images (339 pairs). 120 pairs (30 pairs of each type) were utilized for the validation set with a 256×256 image size [10].

3.1.5. ADNI Datasets

The dataset comprises 2116 subsets of ADNI, including ADNI-1 and ADNI-2. The ADNI1 baseline dataset consists of 1237 subjects' 1.5T T1-weighted MR images, categorized into 389 AD, 432 CN, and 416 MCI subjects. The ADNI-2 baseline dataset includes 879 subjects' 3T T1-weighted MR images [11]. Hao et al. utilized the ADNI dataset comprising 913 patients' imaging data and genetic data, including VBM, FDG-PET, AV45 PET scans, and SNPs data [12].

3.1.6. Multimodal Paired Sets of Brain Images Datasets

The datasets used by Rajalingam et al. comprised six sets of multimodal brain images collected from the Harvard Medical School database and Radiopaedia.org, resized to 256× 256 pixels [13]. Kong et al. utilized six multimodal paired brain images collected from CTisus.com, Harvard University, and the Netherlands TNO image fusion database [14].

3.2. Lung and Respiratory Diseases Datasets

3.2.1. MIMIC-CXR Dataset

The MIMIC-CXR-JPG dataset contains 377,110 chest X-ray images and text reports collected from the Beth Israel Deaconess Medical Center in Boston, MA. It includes 13 disease labels [15]. Upadhyaya et al. utilized the same dataset with 14 diagnostic labels [16]. Moon et al. utilized the publicly available MIMIC-CXR dataset, selecting 91,685 unique AP view image-report pairs [17].

3.2.2. Cough Audio Datasets

Santosh Kumar et al. considered various publicly available cough audio datasets for the diagnosis of COVID-19, including Coswara [18], CoughVid [19], DetectNow [20], and Virufy [21]. The Coswara dataset consists of around 5000 respiratory audio samples collected from nearly 500 participants.

3.2.3. Chest X-ray Datasets

Kumar et al. [22] utilized the multimodal HAIM dataset [23] introduced by Soenksen et al., which integrates chest X-ray images with corresponding electronic medical record (EMR) data derived from the MIMIC-IV clinical database. The HAIM dataset contains approximately 65,000 multimodal records. For their study, Kumar et al. selected 289 patients meeting strict inclusion criteria.

3.2.4. MIMIC-Eye Dataset

The MIMIC-Eye multimodal dataset comprises 670 multimodal instances, combining MIMIC-CXR (chest X-ray images), MIMIC IV-ED (patients' clinical data), and REFLACX (annotations of disease locations in chest X-

rays). The dataset focuses on five different classes [24].

3.2.5. COVID-19 Datasets

The Multimodal COVID-19 (MMC) dataset [25] was employed in the construction and testing of the IRENE clinical diagnosis framework [26]. Liu et al. assessed their medical multimodal large language model using five different COVID-19 datasets: COVIDx-CXR-2 [27], COVID-CXR [28, 29], COVID-19 CT [30], BIMCV-COVID-19 [31], and COVID-HCH

[32].

3.2.6. HAIM-MIMIC-MM Dataset

The HAIM-MIMIC-MM dataset [33] is a comprehensive multimodal clinical dataset created by combining patient information from the MIMIC-IV electronic health records database and the MIMIC-CXR-JPG chest radiography dataset.

3.3. Cancer and Oncology Datasets

3.3.1. SIIM-ISIC Melanoma Classification Dataset

The SIIM-ISIC Melanoma Classification dataset contains 6625 benign and 584 malignant skin lesion images. Text data (clinical notes) are generated using multi-modal models like Google's Gemini Pro 1.5 and OpenAI's GPT-4-turbo [34].

3.3.2. TCGA Dataset

The TCGA dataset has data for three cancer types: BRCA, GBMLGG, and LUSC. It includes 1065 whole slide images (WSIs) contributing about 0.98 TB of data [35]. Dai et al. utilized the TCGA dataset of colorectal patients [36].

3.3.3. NSCLC Dataset

The NSCLC dataset [37] is a multimodal non-small cell lung cancer collection combining radiology, genomic, and clinical data sources.

3.3.4. HAM10000 Dataset

The HAM10000 dataset [38] is a large, multi-source collection of 10,015 dermatoscopic images representing common pigmented skin lesions spanning seven diagnostic categories.

3.4. Cardiovascular Diseases Datasets

3.4.1. Medical Records from Columbia University Irving Medical Center

The dataset was obtained from CUIMC and included over 70,000 patient encounters from January 2013 to August 2018 [39]. The final dataset included 71,589 unique CXRs from 24,689 patients.

3.4.2. Heart Failure and Milder Cardiovascular Diseases Patients' Dataset

This dataset [40] is a consolidated collection of clinical data originally derived from four well-known heart disease databases containing over 1,000 patient records with 14 standardized clinical features.

3.4.3. Intrapartum CTG Dataset

The intrapartum CTG dataset comprises 552 recordings collected from the obstetrics ward of CTU-UHB, Czech Republic. 80 normal and 40 pathological FHR samples were selected [41].

3.5. Infectious Diseases Datasets

3.5.1. SDT6K Dataset

The SDT6K dataset consists of 6250 paired samples of text information and images of swine diseases, including six prevalent infectious swine diseases [42].

3.5.2. MP-IDB Dataset

The MP-IDB has a total of 229 images of Giemsa-stained thin blood smears with four species of malaria parasites [43].

3.5.3. Tuberculosis Patients' Lumbar CT Images Dataset

The dataset contains multimodal image data from 197 patients diagnosed with spinal tuberculosis between 2013 and 2020 [44]. A total of 3,000 lumbar vertebra CT images were randomly selected.

3.6. General Multimodal Datasets

3.6.1. ImageNet-1K Dataset

The ImageNet-1K dataset comprises 1000 image categories and 50,000 validation images.

Multiple OOD datasets are used for testing [45].

3.6.2. Multimodal Dataset of Patients with Ophthalmic Disease

Ma et al. [46] utilized various proprietary datasets (Datasets A–G) for the development, validation, and evaluation of AI models for ophthalmic disease diagnosis.

3.6.3. Multimodal Real Dataset of Patients with Chronic Disease

Chang et al. [47] used multimodal clinical datasets contributed by a hospital for chronic disease diagnosis.

3.6.4. Multimodal Medical Image Datasets

Yao et al. [48] proposed a large-scale dataset incorporating medical images and clinical reports. Pandey et al. [49] proposed a dataset consisting of 30,000 images sourced from public databases.

3.6.5. ACDMD Dataset

The Asian Crohn's Disease Multimodal Dataset contains data from 136 patients [50].

3.6.6. Multimodal Medical Datasets for QNMF

The QNMF project [51] leverages multimodal datasets [52] that fuse imaging information with clinical data.

3.6.7. MIMIC-III Dataset

The MIMIC-III Database [53] is a large publicly available critical care database containing more than 53,000 ICU admissions for over 40,000 patients.

4. Medical Diagnosis With Multimodal Deep Learning

4.1. Overview of Multimodal Medical Diagnosis

Multimodal medical diagnosis is the fusion of heterogeneous medical data, including images, text, electronic health records, and lab results, using a single deep learning model. Single-modality methods that are solely dependent on images or text can fail due to a lack of context. Multimodal deep learning bridges these deficiencies by synergistically integrating the strengths of multiple data sources. Images of medical conditions (X-rays, CT scans, MRI, Ultrasound) offer detailed views into either the structural and functional aspect of patients' conditions, while clinical text data (radiology reports, discharge summaries, and physician narratives) offer semantic information. By combining all these data types, a more comprehensive view of patient health status is represented. Multimodal models also have increased interpretability since attention mechanisms allow direct correlation of particular regions in images to specific text descriptions or clinical variables.

4.2. Common Workflow of Multimodal Medical Diagnosis

The usual multimodal medical diagnosis task involves the following standardized workflow (Fig. 3): (1) data acquisition, (2) preprocessing for each modality (resizing, normalization, segmentation, tokenization, denoising, feature extraction), (3) feature extraction using domain-specific encoders (CNNs, Vision Transformers, LSTMs, BERT-based models), (4) fusion using early, intermediate, late fusion, cross-attention, and/or multimodal transformers, (5) prediction (classification, detection, survival analysis), and (6) evaluation and interpretation using accuracy, AUROC, F1 score, BLEU, C-Index, attention maps, Grad-CAM, SHAP, and prompt traces.

4.3. Sources of Medical Modalities

The main sources of medical modalities can be classified into imaging data, clinical text data, audio signals, and structured clinical data. Table 1 lists the main sources of medical modalities, their corresponding datasets, and applications.

4.4. Description of the Preprocessing Techniques

Various preprocessing techniques have been identified in Table 2. These include sparse coding [54], feature extraction [55], text tokenization with BERT [56], cross attention fusion [57], positional embeddings [58], text standardization [59], patch sampling [60], image compression [61], ROI extraction [62], contrast adjustment [63], data standardization [64], keyword extraction [65], modality alignment via embedding space [66], color normalization [67], masked language modeling [68], self-supervised reconstruction [69], sentence reconstruction [70], terminology normalization [71], dimensionality reduction [72], synthetic data generation [73], report parsing [74], redundant text removal [75], stop word removal and stemming [76], and text concatenation [77].

Table 1
Sources of Medical Modalities.

Modality	Description	Representative Datasets	Applications / Models
Imaging	Radiographic and microscopic imaging (X-ray, CT, MRI, PET, dermatoscopic, histopathological)	MIMIC-CXR [15], HAIMMIMIC-MM [33], TCGA [36],HAM10000[38], ADNI [11, 12]	CNN-based models, Vision Transformers, Hybrid Fusion Models
Textual Clinical Data	Unstructured text reports and structured EHR notes	MIMIC-III [53], MIMICCXR reports [16], SIIM- ISIC [34], ACDMD [50]	LSTM, BERT, Transformers,Con- trastive Learning
AudioSig- nals	Patient-generated audio(cough, breathing)	Coswara [18], CoughVid [19], DetectNow [20], Virufy [21]	CNN-based audio classifiers, Multimodal fusion
Structured Records	Vitalsigns,lab tests,genomic markers	MIMIC-IV [33], HAIM [23], NSCLC [37], Heart Disease [40]	Multi-modal fusion networks, Featurelevel fusion
Multimodal Combinations	Integration of multiple modalities	HAIM-MIMIC-MM[33], MMC[25],Ophthalmic [46]	Multi-modalframeworks, Late/Intermediate Fusion

Table 2
Preprocessing techniques and details.

Technique	Description
Sparse Coding	Compact representation and reconstruction of input data using a few basis functions via L1/L2-regularized optimization [54].
Feature Extraction	Raw data transformation into meaningful representation for model input; reduces dimensionality yet preserves key information [55].
TextTokenization (BERT)	Context-aware tokenization capturing grammatical structure and meaning at word/subword level [56].
CrossAttention Fusion	Attention-based mechanisms to combine image and text features for deep fusion [57].
Positional Embeddings	Encoding order of tokens or image patches to preserve sequence structure [58].
TextStandardization	Unifying formats, casing, and symbols while removing inconsistencies [59].
Patch Sampling	Breaking large images into smaller fixed-size patches for efficient transformer processing [60].
Image Compression	Reducing image size while retaining key visual features [61].
Technique	Description
ROI Extraction	Identifying diagnostically relevant content regions and removing irrelevant areas [62].
ContrastAdjustment	Enhancing visibility of structures by modifying the dynamic range of images [63].
Data Standardization	Ensuring consistent statistical properties (zero mean, unit variance) for numerical stability [64].
KeywordExtraction	Identifying relevant clinical terms reflecting key diagnostic features [65].
ModalityAlignment	Mapping text, image, and other modalities into a common embedding space [66].
ColorNormalization	Aligning color distributions across datasets to reduce staining variations [67].
MaskedLanguage Modeling	Masking portions of text for contextual understanding and cross-modal learning [68].
Self-Supervised Reconstruction	Reconstructing missing or transformed parts of data without labeled supervision [69].

Technique	Description
SentenceRecon- struction	Fixing grammar and structure in unorganized clinical text using NLP [70].
TerminologyNormalization	Connecting clinical expressions to standardized ontologies (e.g., SNOMED-CT) [71].
Dimensionality Reduction	Compressing high-dimensional data using PCA or UMAP while retaining key features [72].
SyntheticData Generation	Creating artificial data to augment training using GANs or language models [73].
Report Parsing	Extracting structured information from medical reports [74].
RedundantText Removal	Removing repeated or irrelevant segments of clinical reports [75].
StopWordRe- moval & Stemming	Removing non-informative words and reducing words to root forms [76].
TextConcatena- tion	Merging multiple related text fields into unified strings [77].

4.5. Model Architectures

Table 3 outlines representative multimodal architectures designed for various medical tasks and modality combinations.

Amongst all the reviewed papers, lightweight CNNs outperform in efficiency, while transformerbased and cross-attention architectures provide deeper semantic correspondences between modalities. Table 4 provides comparative observations across models.

Table 3: Summary of model architectures.

Model	Input Modalities	Architecture	Ref.
MCX-Net	CXR images + clinical text	Conv. layers + Transformer	[15]
AMMDMLF	Cough audio	Deep audio extraction + fusion	[78]
Late Fusion	CXR + clinical text	Separate pipelines, late fusion	[16]
MDF-Net	Eye images + clinical data	Multi-modal deep feature network	[24]
IRENE	CXR, CT, lab tests	Unified model, attention mechanisms	[26]
Med-MLLM	COVID-19 images + text	Multi-modal language-image model	[79]
CBP-MMFN DHP-MMFN	/ CXR + clinical text	Cross-modal attention fusion	[80]
Deep U-Net + Darknet-19	CXR + cough audio	Hybrid CNN models	[81]
MedViLL	CXR + report text	Transformer-based vision-language	[17]
HAIM + XG- Boost	CXR + structured EHR	Multi-modal fusion + boosting	[23]
IOMIDS	Ophthalmic images + dialogs	NLP and vision fusion	[82]
Hybrid CNN + DT	Multi-modal images	CNN features + decision tree	[49]
QNMF	Images + EHR + blood tests	Quantum-inspired network	[51]
Label-aware CNN+LSTM	MIMIC-III chronic data	Temporal-clinical note fusion	[83]

Table 4: Comparative observations across models.

Key Observations	
vs.	QNMF and hybrid CNN+DT are lightweight; BMFL, MDLM, IOMIDS are heavy due to cross-attention and LLM components.
Real-Time Feasibility	Hybrid CNN+DT and Combined Fusion suit real-time deployment; BMFL and IOMIDS are resource-intensive.
Trustworthiness	Models with XAI (QNMF, MDLM) provide clearer explanations; absence of XAI raises black-box concerns.
Data Requirements	BMFL requires fMRI (impractical); IOMIDS needs large dialog datasets; QNMF requires less data.
Fusion Strategies	Early fusion (QNMF) improves speed; late fusion simplifies structure; crossattention enhances coupling but is costly.
Robustness	QNMF shows noise resistance; MDLM risks overfitting; BMFL suffers inter-subject variability.
Generalizability	Combined Fusion and QNMF generalize best; dataset-anchored models show limited transferability.

5. Results Analysis

5.1. Overview of Included Studies

The cited literature covers a wide range of clinical areas, all published between 2021 and 2025. The studies include varied data modalities including medical imaging (CT, MRI, X-ray, fundus), clinical text (EHR entries, diagnostic texts, patient conversations), tabular data, biosignals, and audio-based respiratory signals. Multimodal fusion appears as a crucial aspect, with models using imaging, clinical texts, and EHR data for improved diagnostic performance.

5.2. Dataset Diversity Across Studies

Among the reviewed studies, data sources include medical image analysis, text-based clinical data, lab results, and physiological signals. Table 5 summarizes the comparison of dataset types.

Table 5
Comparison of dataset types across reviewed studies.

Dataset Type	Examples	Ease of Use	Main Challenges	Best Models	Strengths	Weaknesses
Medical Imaging	QNMF, Ma et al., Yao et al.	Moderate	Scanner variability, noise, ViTs	CNNs, segmentation	Rich spatial information	Large storage; info variability
Clinical Text	MIMIC-III, IOMIDS	Low	Abbreviations, ambiguity	BERT, GPT, LSTM	Captures heavy reasoning	Preprocessing
Structured EHR	MIMIC-III, QNMF	High	Missing values, irregular sampling	LSTMs, XGBoost	Easy to integrate	Limited to inter-richness
fMRI/Signals	Zhao et al.	Very Low	Inter-subject variability	Fusion networks	Cognitive data	Noisy, small
Multi-Dialog	IOMIDS	Low–Mod.	Text-image alignment	Multimodal transformers	Realistic workflow	Hard to standardize
Multisource	QNMF, MDLM	High Complexity	Alignment, Cross weighting	attention	Most informative	Computationally demanding

5.3. Performance Comparison Across Disease Domains

Cancer and chronic disease models generally achieve the highest accuracies (92–97%). Ophthalmic and dermatology systems exhibit more moderate performance ($\approx$ 79–85%). Respiratory and cardiovascular models display mixed performance profiles.

Tables 6–11 present detailed analyses of research articles organized by disease domain.

Table 6: Analysis of research articles on Brain and Neurological Disorders (2021–2025).

Ref.	Dataset	Pre- Processing	Model	Results	XAI	Limitations
Li et al., 2024 [84]	Parkinson's dataset	RGB encoding, Sparse encoding, Feature extraction	MTL framework, STN	Acc: 95.08%, Prec: 89.13%	N/A	Limited dataset; fusion challenges; computational complexity
Dentamaro et al., 2024 [7]	PPMI	Missing value replacement, Onehot encoding, Normalization	DenseNet + Excitation	Acc: 96.6%, AUC: 98.9%	Integrated Gradients, Attention Heatmap	Overfitting risk; interpretability challenges
Wang et al., 2024 [8]	TADPOLE, ABIDE	Normalization, Imputation, Dimensionality reduction	HMGL	Acc: 93.14%, AUC: 94.17%	N/A	Limited diversity; scalability issues
Lu et al., 2024 [9]	BPPV	Video segment selection, Frame sampling	BKTDN	Acc: 81.7%, Sens: 94.1%	N/A	Limited dataset; missing records
Ouyang et al., 2023 [85]	Cervical spine data	Data quality checks, Missing data handling	XGBoost, Clinical BERT, ConvNeXt	Acc: 91.6%, F1: 92%	GradCAM, SHAP	Textual quality issues; limited validation
Yang et al., 2022 [10]	Whole Brain Atlas	YCbCr conversion, Luminance channel fusion	HPN	AG: 0.819, CC: 0.869	N/A	Limited datasets; suboptimal CTMRI fusion
Hao et al., 2022 [12]	ADNI (imaging + genetic)	VBM extraction, PET registration, SNP encoding	MSLPL	Acc: 95.14%, AUC: 95%	N/A	Lack of consistency constraints; limited size

Ref.	Dataset	Pre- Processing	Model	Results	XAI	Limitations
Pei et al., 2022 [11]	ADNI-1, ADNI-2	CAT 12/SPM, Skull stripping, Affine registration	PKG-Net	Acc: 97.28%	N/A	MRI-specific; scanner variability unaddressed
Rajalingam et al., 2022 [13]	CT, MRI, SPECT, PET	Image resizing, NSCT decomposition	NSCTGIF	mSSIM: 98.52%, PSNR: 55.06 dB	N/A	High computational cost; fusion artifacts
Kong et al., 2022 [14]	Brain im- ages	Grayscale conversion, RGB to YUV	CELM	QMI: 1.2634	N/A	Limited datasets; weak ELM repre- sentation

Table 7: Analysis of research articles on Lung and Respiratory Diseases (2021–2025).

Ref.	Dataset	Pre-Processing	Model	Results	XAI	Limitations
Yang et al., 2025 [15]	MIMIC-CXR-JPG	Resizing, Normalization, Patching, Tokenization	MCX-Net	Macro F1: 45.1%, AUROC: 81.6%	Attention maps, Grad-CAM	Dataset specificity; text quality dependence
Pandey et al., 2024 [78]	Cough audio	CLAHE, Cropping, Normalization	AMMDMLF	Acc: 92.5%, Prec: 91.8%	N/A	Limited dataset; class imbalance; overfitting
Upadhyay et al., 2024 [16]	MIMIC CXR	Resizing, Tokenization, Augmentation	Late Fusion	Acc: 94.2%	N/A	Limited generalizability; class imbalance
Kumar et al., 2023 [22]	CXR + clinical data	Resizing, Augmentation, Synthetic generation	Late/Intermediate fusion	Acc: 93.15%, F1: 94.75%	N/A	Limited size; crossmodality imbalance
Hsieh et al., 2023 [24]	MIMICEye	Resizing, Clinical data spatialization	MDF-Net	Avg Prec: 31.69%	N/A	Small dataset; limited explainability
Zhou et al., 2023 [26]	CXR, MMC	Resizing, Cropping, Augmentation, Tokenization	IRENE	AUROC: 92.4%	Attention map, Grad-CAM	Limited diversity; modal deficiency
Liu et al., 2023 [79]	COVID-19 datasets	Masked LM, Sentence reconstruction, Contrastive learning	Med-MLLM	Diag Acc: 98.7%, BLEU-4: 63.2%	N/A	Limited labeled data; multilanguage challenges

Ref.	Dataset Pre-Processing		Model	Results	XAI	Limitations
Shetty et al., 2023 [80]	Indiana/KMCRotation, Stem-CXR ming, Tokenization		CBPMMFN, DHPMMFN	Acc: 98.88%, AUROC: 98.89%	N/A	Small dataset; high execution time
Kumar et al., 2022 [81]	CXR + Coswara cough	Resizing, Lung segmentation, Noise removal	Deep U-Net, Darknet-19	Overall Acc: 92.03%	N/A	Unmatched CXRcough samples
Moon et al., 2022 [17]	MIMIC-CXR, Open-I	Resizing, Report parsing, WordPiece tokenization	MedViLL	AUROC: 98.0%, F1: 83.9%	Attention visualization	Single view; data imbalance
Soenksen et al., 2022 [23]	HAIMMIMIC-MM	Normalization, Resizing, Text concatenation	HAIM + XGBoost	Δ AUROC: 6–33%	Shapley values	Heterogeneity; error propagation

Table 8: Analysis of research articles on Cancer and Oncology (2021–2025).

Ref.	Dataset	Pre- Processing	Model	Results	XAI	Limitations
Chakkarapan et al., 2025 [34]	SIIM-ISIC	Resizing, Color/contrast normalization, Text standard-ization	BERT + ResNet-50, ALBEF	Acc: 99.51%, F1: 97.03%	N/A	Dataset imbalance; synthetic note dependence
Dai et al., 2024 [36]	TCGA	Feature removal, ROI extraction, Graph construction	MLCSP	C-Index: 0.736	SHAP, Cell Graph Heatmap	Limited dataset; high computational cost
Li et al., 2024 [35]	TCGA	Patch sampling, Self-supervised reconstruction	PMFNSSL	Acc: 96.1%, F1: 90.9%	GradCAM, IG	Limited annotations; spatiotemporal incongruity
Vanguri et al., 2022 [37]	NSCLC	CT segmentation, Radiomics extraction	DyAM	Acc: 80% (CI: 0.74–0.86)	Attention mechanism	Restricted cohort; annotation variability
Wang et al., 2021 [86]	HAM10000	Imputation, Hair removal, Inpainting, Augmentation	IM-CNN	Acc: 95.1%, AUC: 97.8%	GradCAM, SHAP	Interpretability–accuracy trade-off; noise

Table 9: Analysis of research articles on Cardiovascular Diseases (2021–2025).

Ref.	Dataset	Pre- Processing	Model	Results	XAI	Limitations
Wang et al., 2024 [87]	CUIMC records	Filtration, Temporal grouping, SMOTE, Resizing	VAE	Acc: 99.58%, F1: 99.38%	N/A	Limited dataset; potential biases
Liu et al., 2024 [40]	Heart disease dataset	ECG binarization, Denoising, Normalization, Augmentation	Multimodal Fusion	Feature Fusion Acc: 99.92%	N/A	Limited data; augmentation dependence
Zhang et al., 2022 [41]	Intrapartum CTG	Missing data filling, GADF conversion, Augmentation	MMIF: CCPViT + MRAN	Acc: 96.3%, AUC: 96.2%	N/A	Unaligned data; modality sensitivity

Table 10: Analysis of research articles on Infectious Diseases (2021–2025).

Ref.	Dataset	Pre- Processing	Model	Results	XAI	Limitations
Yang et al., 2025 [42]	SDT6K	Stop-word removal, EDA augmentation, SSIM deduplication	TGFN-SD	Acc: 94.48%, F1: 94.40%	GradCAM, Attention weights	Image variability; phenotypic overlap
Chen et al., 2023 [43]	MP-IDB	ACE, Augmentation, Resizing	SDU-Net, MobileNet V1	DICE: 98.47%, Acc: 98.83%	N/A	Poor scalability; limited validation
Li et al., 2022 [44]	TB lumbar CT	Window adjustment, Watershed segmentation	Multimodal fusion CNN	Acc: 98.33%, AUC: 99.84%	Heatmap, Grad-CAM++	Small dataset; overfitting risk

Table 11: Analysis of research articles on General Multimodal Frameworks (2021–2025).

Ref.	Dataset	Pre-Processing	Model	Results	XAI	Limitations
Zhao et al., 2025 [45]	ImageNet1K	fMRI prediction, Patchification, Cross-attention	BMFL	Top-1 Acc: 85.72%	N/A	fMRI variability; high collection costs
Ma et al., 2025 [82]	Ophthalmic data	Dialog curation, YOLOv7 segmentation, Augmentation	IOMIDS	Internal Acc: 79.6%, Triage: 88.7%	CAM, Prompt traces	Lower accuracy for certain conditions
Chang et al., 2024 [47]	Chronic disease data	Standardization, Denoising, Feature fusion	Combined Fusion	Acc: 92.4%, F1: 92.3%	N/A	N/A
Yao et al., 2024 [48]	Medical image dataset	Standardization, Stop words removal, Stemming	MDLM: CNN + Bi-LSTM + Attention	Acc: 96.42%, Recall: 98.48%	N/A	Limited scope; potential overfitting
Pandey et al., 2024 [49]	Medical image dataset	Standardization, Median filtering, Augmentation	CNN (ResNet-34) + Decision Tree	Acc: 93.4%, F1: 92.3%	N/A	Limited diversity; black-box CNNs
Jia et al., 2024 [50]	ACDMD denoising, HSV normalization	Noise2Noise	MICDnet	Acc: 85.0%	N/A	Paucity of data; rudimentary fusion
Qu et al., 2023 [51]	MedMNIST, Resizing, AnWisconsin, gle encoding, COVID-19 extraction, Early fusion	QCNN	QNMF: QCNN + VQC	Acc: 97.61%, F1: 97.45%	CAM, Prompt traces	Limited quantum resources; NISQ constraints

Ref.	Dataset	Pre- Processing	Model		Results	XAI	Limitations
Yang et al., 2021 [83]	MIMIC-III	Window selection, Word2Vec, Zero padding	Labelaware CNN LSTM	+	AUCROC: 0.875	Attention scores	Imbalanced dataset; gap with clinical diagnosis

5.4. Critical Assessment of Reported Performance

Several studies in this review report near-perfect accuracy values (e.g., 99.92% for cardiovascular fusion, 99.58% for VAE-based cardiac detection, 99.51% for melanoma classification). While these results appear promising, they warrant critical scrutiny. First, many of these high-accuracy studies rely on small, institution-specific datasets without external validation, raising concerns about overfitting to training distributions. Second, data leakage risks are insufficiently addressed in several works; for example, temporal overlap between training and test splits, or patient-level leakage where multiple samples from the same patient appear in both sets, can inflate reported metrics. Third, aggressive data augmentation techniques such as SMOTE, which are commonly employed in cardiovascular and oncology studies, can artificially boost performance on minority classes without genuinely improving the model’s discriminative ability on unseen data. Fourth, the absence of calibration analysis in most studies means that high accuracy does not necessarily translate to clinically reliable probability estimates. Finally, cross-study comparison is fundamentally hindered by the use of different evaluation protocols, dataset splits, and preprocessing pipelines. Studies reporting AUROC above 98% on small institutional datasets should be interpreted cautiously, as these results are unlikely to generalize to multi-center clinical deployment without significant performance degradation.

5.5. Quantitative Trends Across Reviewed Studies

Figure 4 summarizes the key quantitative patterns identified across the 40 reviewed studies. In terms of architectural trends, CNN-based and late fusion models dominate the earlier period (2021–2022), comprising approximately 65% of reviewed architectures, while transformer-based and cross-attention models have become increasingly prevalent from 2023 onward, accounting for roughly 55% of studies published in 2024–2025 (Fig. 4b). Regarding modality usage, imaging remains the most common modality, present in 38 out of 40 studies (95%), followed by clinical text or EHR data (72.5%), audio signals (7.5%), and genomic data (7.5%) (Fig. 4a). Explainability integration remains inconsistent: only 15 out of 40 studies (37.5%) incorporate any form of XAI technique, with Grad-CAM being the most frequently used (appearing in 9 studies), followed by SHAP (5 studies) and attention visualization (6 studies) (Fig. 4c). Performance varies substantially by disease domain: neurological disorder models report a median accuracy of approximately 95%, oncology models around 96%, cardiovascular models above 96%, and infectious disease models around 98%, while general multimodal frameworks show the widest variance (85–97%). However, higher reported accuracy tends to correlate with smaller dataset sizes and single-institution evaluation, suggesting that performance inflation is a systematic concern across the field. Notably, none of the 40 reviewed studies conducted

prospective clinical validation, and only 3 studies employed multi-institutional test sets, underscoring a significant gap between reported research performance and clinical readiness.

6. Challenges and Future Research Directions

6.1. Class Imbalance in Multimodal Datasets

The problem of class imbalance occurs when there are greatly fewer samples in one class compared to the other classes [88]. In the medical context, this bias prevents multimodal systems from correctly detecting rare and serious medical issues [89]. Data-level solutions include SMOTE [88, 90], while algorithm-level techniques incorporate specialized loss functions such as Focal Loss [91].

6.2. Limited Clinical Metadata Integration

Most existing multimodal models insufficiently use clinical metadata including demographic variables, medical history, comorbidities, and lab results [92, 93]. Advanced fusion methodologies combining diagnostic information with clinical metadata [94] and attention mechanisms [95] are being developed to address this.

6.3. Modal Deficiency Handling

Modal deficiency occurs when a patient has one or more missing modalities of information [96]. Strategies include masked modeling, data imputation using VAEs or GANs [97], and transformer models that dynamically adapt to available modalities [98].

6.4. High Computational Cost and Complexity

Deep learning models involving multiple modes involve significant computational complexity. CNNs for images and Bi-LSTMs for text [99] create heavy computational loads. Solutions include efficient architectures (e.g., ResNet over VGG) [100, 101] and model compression techniques. Deep reinforcement learning approaches also show promise [102, 103].

6.5. Limited Quantum Computing Resources

Quantum neural networks are impeded by NISQ-era hardware limitations [104, 105]. Hybrid classical-quantum models [106], variational quantum algorithms [105], and quantum error mitigation methods [107] are being developed to address these constraints.

6.6. Phenotypic Overlap

Several diseases can have similar signs and symptoms, leading to diagnostic confusion [108, 109]. Multitask learning and contrastive learning algorithms [110] are being developed to improve discrimination between diseases with overlapping phenotypes.

6.7. Limited Multimodal Dataset Size

Deep learning models require large datasets, but multimodal medical datasets are often small due to privacy issues, annotation costs, and disease rarity [111, 112, 113, 114]. Data augmentation, synthetic data, transfer

learning, and regularization [115] are proposed to mitigate this limitation. Self-supervised representation learning approaches, such as SimCLR-based methods applied to breast cancer classification [4] and multi-class brain tumor detection [5], have shown that competitive diagnostic performance can be achieved on smaller datasets by learning robust feature representations without extensive labeled data.

6.8. Heterogeneous Modality Fusion Complexity

Heterogeneous modality fusion faces challenges including misalignment and modality gap [116], dimensionality and scale disparity [117], missing modalities [118], and computational complexity. Countermeasures include modality-specific encoders, cross-modal attention, contrastive learning, and gate/attention mechanisms [119, 120].

6.9. Deep Learning Model Interpretability Challenges

Deep learning models provide predictions without explaining the decision-making process [121, 122, 123]. Model-agnostic methods (SHAP, LIME) and intrinsic methods (attention mechanisms) have been proposed [124], but effectively balancing interpretability and accuracy remains challenging. Recent work has explored combining XAI with domain-specific models, including interpretable risk estimation for cardiovascular disease [1], GradCAM-based visualization for gallbladder disease classification [6], and EDA-SimCLR-SHAP pipelines that demonstrate how self-supervised paradigms can provide competitive accuracy alongside superior interpretability [3].

6.10. Trade-off Between Interpretability and Accuracy

Very transparent models cannot deliver needed accuracy in complex clinical tasks, while very accurate models are opaque [125]. Hybrid approaches including post-hoc XAI techniques and inherently interpretable architectures are being investigated [128]. For instance, interpretable 1D-CNN frameworks for symptom-based lung cancer prediction have demonstrated that transparent attribution mechanisms can be integrated without substantial accuracy loss

[2].

6.11. Spatio-Temporal Incongruity in Multimodal Data Fusion

Multimodal data components may not be aligned in space and time [129, 130]. This creates ambiguity in fusion [131, 132] and can confuse disease progression with modality differences [133]. Temporal filtering, spatial registration, and attention-driven weighting are needed.

6.12. Asynchronous Data Alignment Issue

Multimodal data captured at different times or irregular intervals undermines cross-modal associations [134, 135, 136]. Temporal windowing, interpolation, attention-based alignment layers, and time-warping techniques [137, 138] are proposed solutions.

6.13. Limitations of This Review

This review has several limitations that should be acknowledged. First, the inclusion of 40 studies, while sufficient for a focused survey, does not capture the full breadth of multimodal medical AI research, and the

selection may reflect publication bias toward studies reporting positive results. Second, a formal meta-analysis was not conducted due to the heterogeneity of tasks, metrics, datasets, and evaluation protocols across the reviewed studies, which limits the strength of quantitative comparisons. Third, although PRISMA guidelines were followed, a standardized quality scoring tool such as QUADAS-2 was not applied, and inter-reviewer agreement was not formally assessed, which may introduce subjectivity in study selection and evaluation. Fourth, the review is limited to English-language publications, potentially excluding relevant work published in other languages. Fifth, the rapid pace of development in this field means that studies published after the search period (post-2025) are not captured. Finally, the six disease domain categories used for organization, while practical, impose boundaries that may oversimplify the cross-domain applicability of certain general-purpose multimodal frameworks.

7. Conclusions

This review demonstrates the maturity of multimodal deep learning and its role as a prominent approach in medical diagnosis across a range of disease types. From 2021 to 2025, there is a clear trend that increased diversity and multimodality are highly beneficial for improving diagnostic accuracy, though also quite challenging.

Imaging datasets were the most rich in spatial information, text provided depth of context, and EHR supplied medical detail via structured data, confirming that no single modality is globally dominant and that multimodal fusion generally outperforms single-modality approaches in the majority of reviewed studies, though the magnitude of improvement varies considerably across disease domains and data settings. Preprocessing and augmentation methods had a substantial effect on robustness.

Attention models and Transformer models (MedViLL, IRENE, ALBEF) were found to be more accurate with better AUROC values, particularly for complex multimodal fusion tasks, but required more computational power. Late fusion models offered the best compromise between efficiency and application. Multimodal models generally outperformed unimodal models by 3–25% in accuracy, AUC, and F1.

Despite achievements, important bottlenecks persist: small dataset size, modality variability, data misalignment, spatio-temporal misalignment, low interpretability, and clinical generalization. Future research should address multimodal dataset standardization, alignment techniques, efficient architectures, and trustworthy explainable AI to ensure smooth transition from research to clinical practice.

Abbreviations

A comprehensive list of abbreviations is provided in Appendix Appendix A.

Appendix A. Abbreviations

Full Form	Abbreviation
Explainable Artificial Intelligence	XAI
Multi-Task Learning	MTL
Spatial-Temporal Network	STN
Root Mean Square Error	RMSE
Mean Absolute Error	MAE
Parkinson's Progression Markers Initiative	PPMI
Area under the ROC Curve	AUC
Alzheimer's Disease Prediction Of Longitudinal Evolution	TADPOLE
Benign Paroxysmal Positional Vertigo	BPPV
SHapley Additive exPlanations	SHAP
Alzheimer's Disease Neuroimaging Initiative	ADNI
Voxel-Based Morphometry	VBM
Single Nucleotide Polymorphisms	SNP
Positron Emission Tomography	PET
Convolutional Neural Network	CNN
Medical Information Mart for Intensive Care – Chest X-ray	MIMIC-CXR
Area Under the ROC Curve	AUROC
The Cancer Genome Atlas	TCGA
Whole Slide Images	WSI
Integrated Gradients	IG
Synthetic Minority Oversampling Technique	SMOTE
Electrocardiography	ECG
Functional Magnetic Resonance Imaging	fMRI
Bidirectional Long Short-Term Memory Networks	Bi-LSTM
Quantum Neural Network-based Multimodal Fusion System	QNMF
Long Short-Term Memory	LSTM
Vision Transformer	ViT
Generative Pre-trained Transformer	GPT
Multi-Layer Perceptron	MLP

Variational Autoencoder	VAE
Contrastive Language–Image Pretraining	CLIP
Principal Component Analysis	PCA
Magnetic Resonance Imaging	MRI
Computed Tomography	CT
Region of Interest	ROI
Electronic Health Records	EHR
Natural Language Processing	NLP
Full Form	Abbreviation
Early Fusion (Feature-level fusion)	EF
Late Fusion (Decision-level fusion)	LF
Cross-Modal Attention	CMA

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Figures

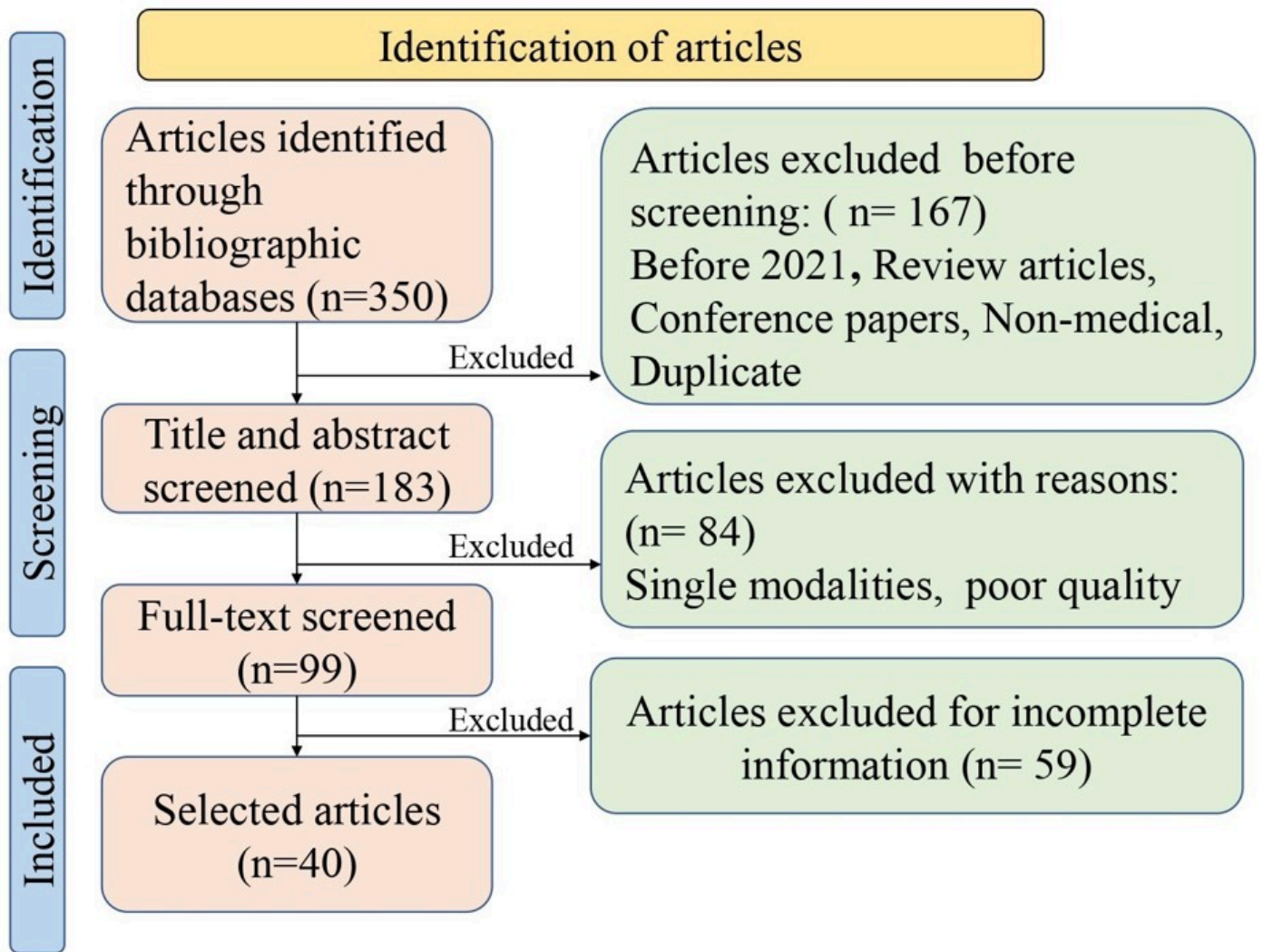


Figure 1

PRISMA flow diagram of the article selection procedure.

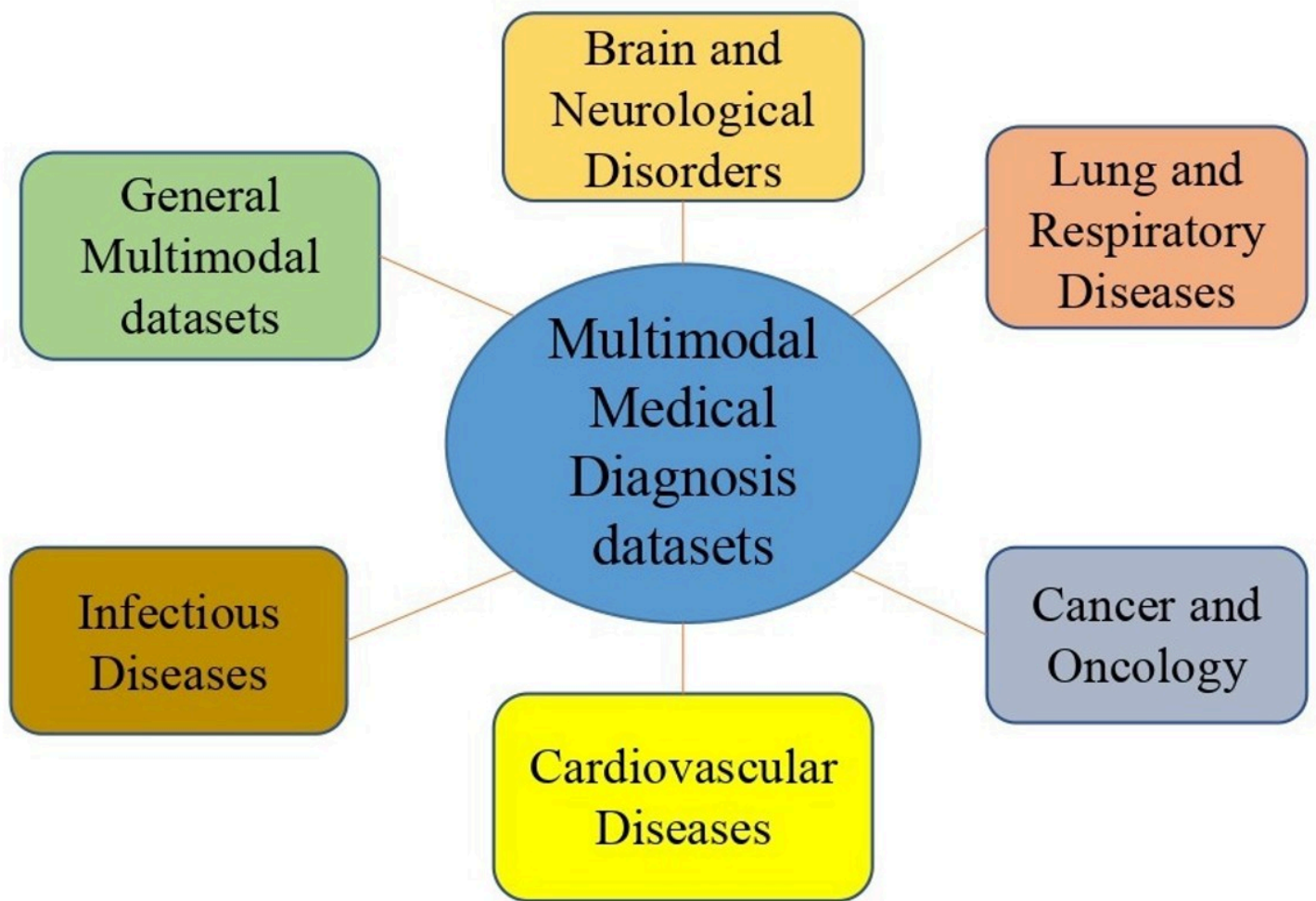


Figure 2

Utilized datasets in multimodal learning for medical diagnosis.

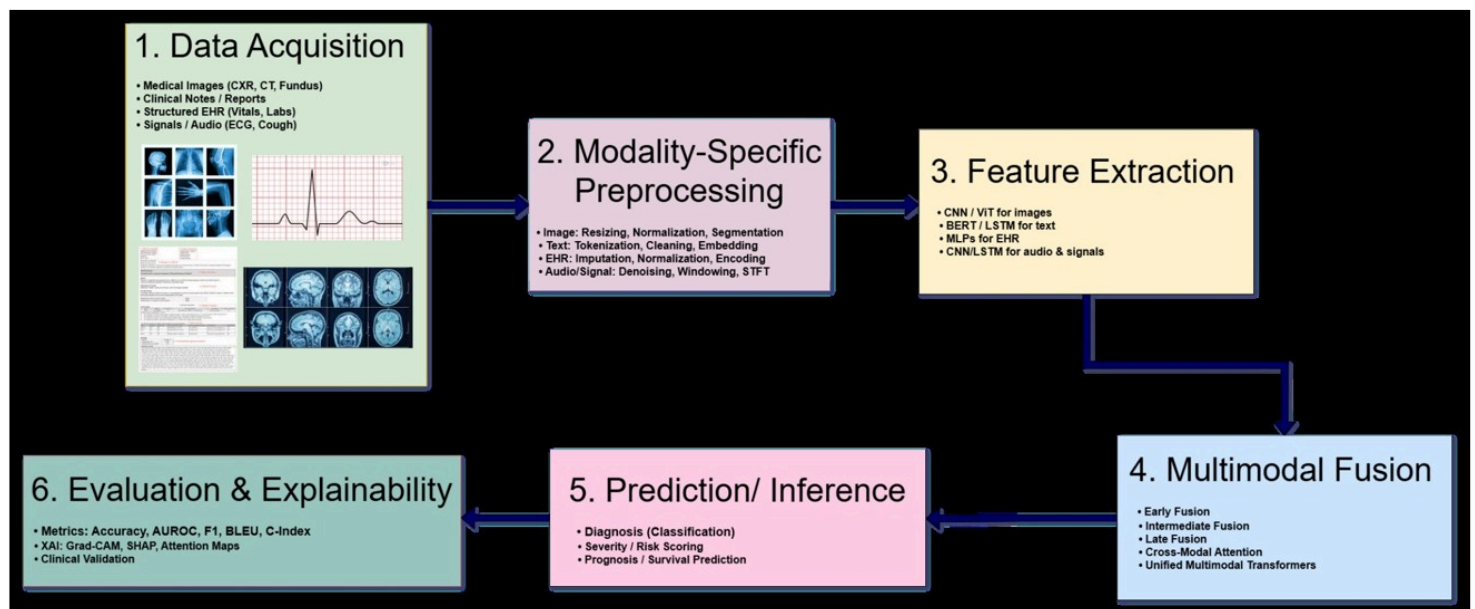


Figure 3

Common workflow of multimodal deep learning in medical diagnosis.

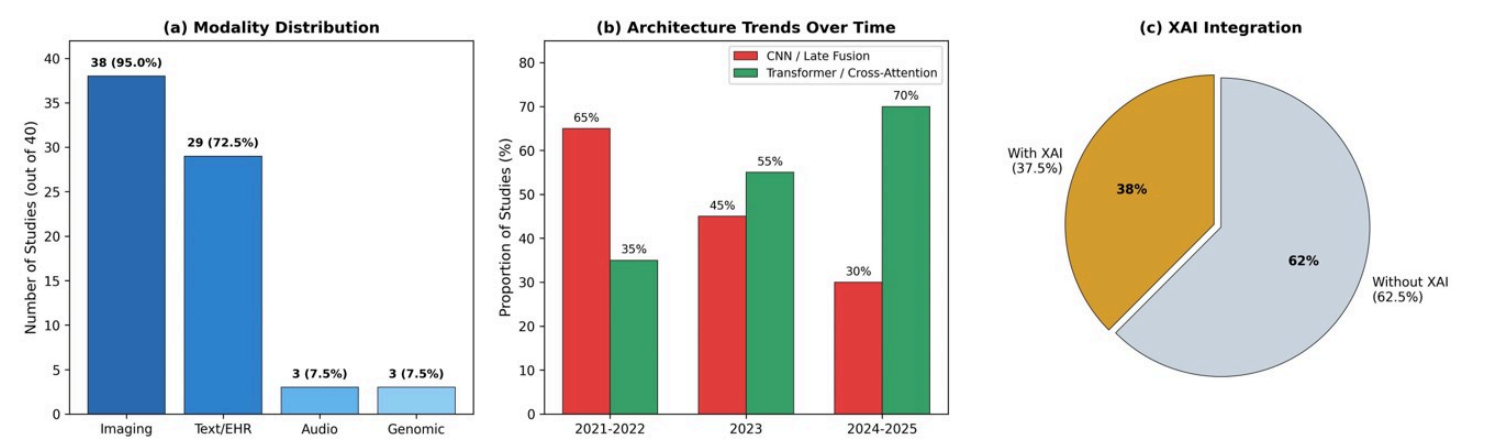


Figure 4

Quantitative trends across 40 reviewed multimodal medical diagnosis studies: (a) distribution of data modalities, (b) architectural trends from CNN/late fusion toward transformer/cross-attention models over time, and (c) proportion of studies incorporating explainable AI techniques.